
```

% Ezinne Akabueze, CMOR 220, Spring 2026, Project 4: Population
% Project4_Population.m
% This coding program is tasked to use Eulers method then the ODE45 command
% to approximate ordinary differenrential equations' solutions.
% Last modified: 3/3/26
clc;
clear;

disp('Project 4')
% Defining all necessary functions for Project 3:

% Function for Euler's Method, where we try to approximate the solution
function [a,t0] = Eulers(chng);
    t0 = 0:chng:4;
    a(1) = 8;
    for k=1:length(t0)-1
        a(k+1) = a(k)+chng*(2*a(k)*t0(k));
    end
end

% Task 1: solve the logistic growth model for the rabbit population in the
absence of predators.
% Set initial conditions
disp('Task 1')
K = 100;
chng = 0.01;
r = 1;
f = @(t, R) R*(r-R/K); % Define the logoistic growth model

% Task 2: solve the predator-prey problem described in Section 3
disp('Task 2')
% Set initial conditions
k1 = 3;
k2 = 3e-3;
k3 = 6e-4;
k4 = 0.5;
R0 = 1000;
F0 = 500;

% Task 3: create figure 1 for parts RFa and RFb
disp('Task 3')
K = 100
T = 5
r = 1
figure(1)
plot(T, r) % plot population vs time
title('T = 5, K = 100, r = 1')
xlabel('time')

```

```

ylabel('population')

% Task 4: create figure 2, using the rabbit population for your x-variable
and the fox population for your y-variable.
disp('Task 4')

figure(2)
plot(T, r) % plot fox population vs rabbit population
title('fox population vs rabbit population')
xlabel('rabbit population')
ylabel('fox population')

% Task 5: create movie
disp('Task 1')

v = VideoWriter('foxAndRabbit');
open(v); % open video
f = getframe(gcf); % Add current frame f to video v
hold off;
close(v); % close video

Project 4
Task 1
Task 2
Task 3

K =

    100

T =

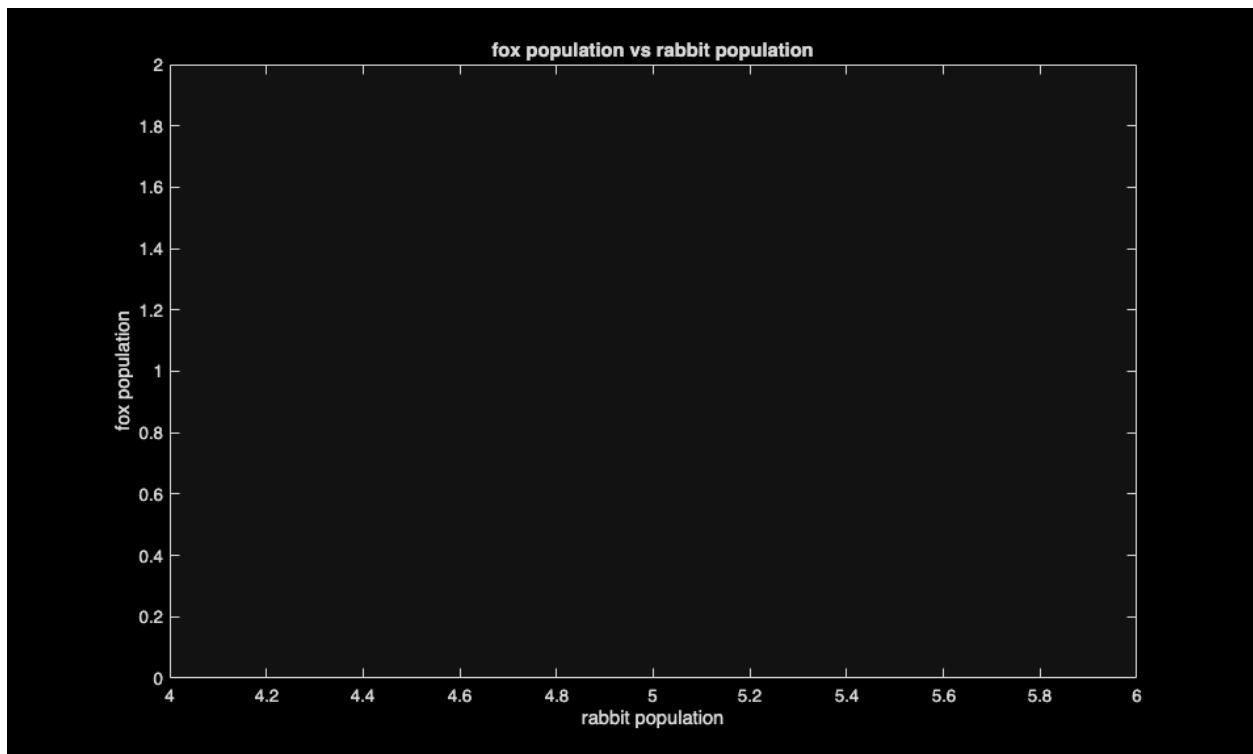
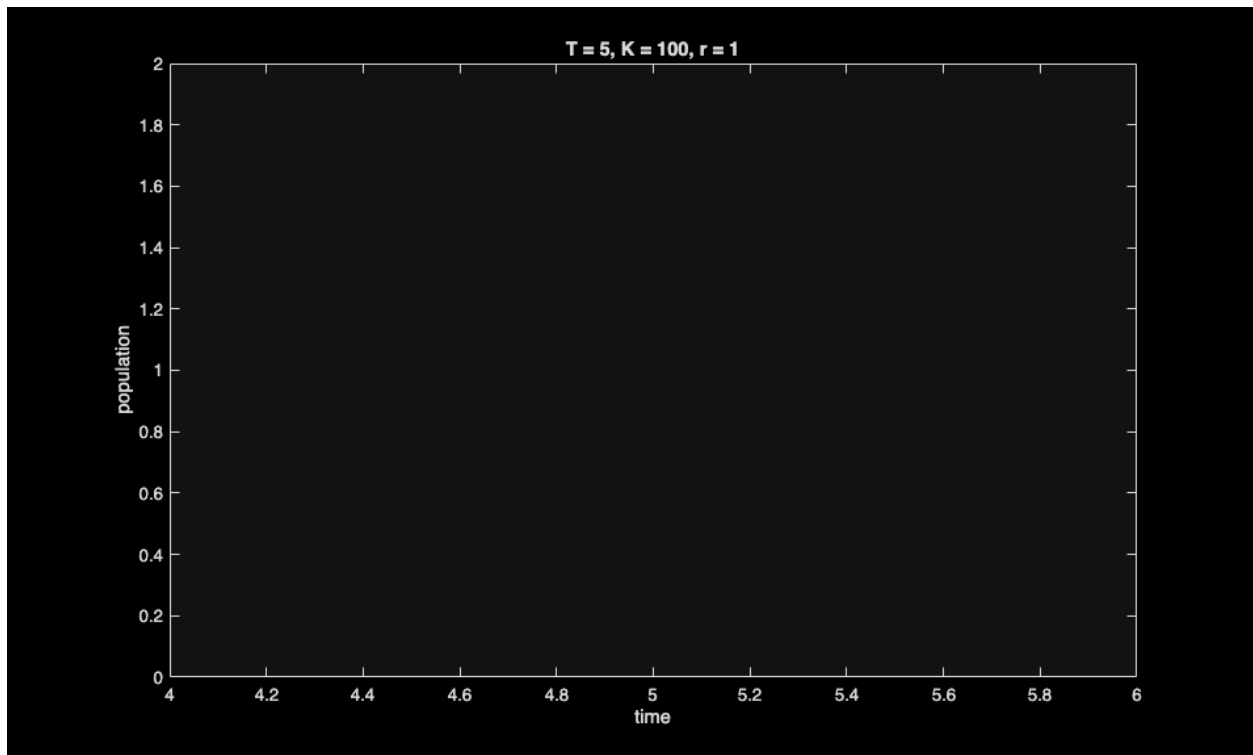
     5

r =

     1

Task 4
Task 1
Warning: No video frames were written to this file. The file may be
invalid.

```



Published with MATLAB® R2025b